



TechBiz Academy

Week 6 · Task 17

Password Cracking — Hashes (John / Hashcat)

Submitted By

Ali Hamza

Registration No

TBS-2026-017

Course

Ethical Hacking Internship

Date

07 August 2026

Organization

Tech Biz Security

Table of Contents

Sr. No.	Contents	Page No.
1	Student Information	1
2	Objective	1
3	Tools Used	2
4	Resources	2
5	Task 1 – Study the Lab (TryHackMe)	3
6	Task 2 – Create a Sample MD5 Hash	4
7	Task 3 – Save the Hash into a File	5
8	Task 4 – Recover Password Using CrackStation	6
9	Task 5 – Perform a Wordlist Attack Using John the Ripper	7
10	Task 6 – Wordlist Attack Result	8
11	Task 7 – Verify the Cracked Password	9
12	Task 8 – Hashcat Reference	10
13	Difference Between Hashing and Encryption	11
14	Results	12
15	Conclusion	12
16	References	13
17	Appendix A – Hash Cracker Badge	14
18	Screenshot Placement Summary	15

Objective

The objective of this lab was to learn how password hashing works by generating an MD5 hash, storing it in a file, recovering the password using both an online hash lookup (CrackStation) and an offline dictionary attack (John the Ripper), verifying the recovered password, and understanding the difference between hashing and encryption.

Tools Used

- Kali Linux
 - MD5
 - John the Ripper
 - RockYou Wordlist
 - CrackStation
 - TryHackMe
 - Hashcat Wiki
-

Resources

- TryHackMe — Crack the Hash (Free Room)
 - Hashcat Wiki (Reference)
 - Video: "John the Ripper hashcat tutorial beginners"
 - Video: "hashing vs encryption explained"
-

Task 1 – Study the Lab

Objective

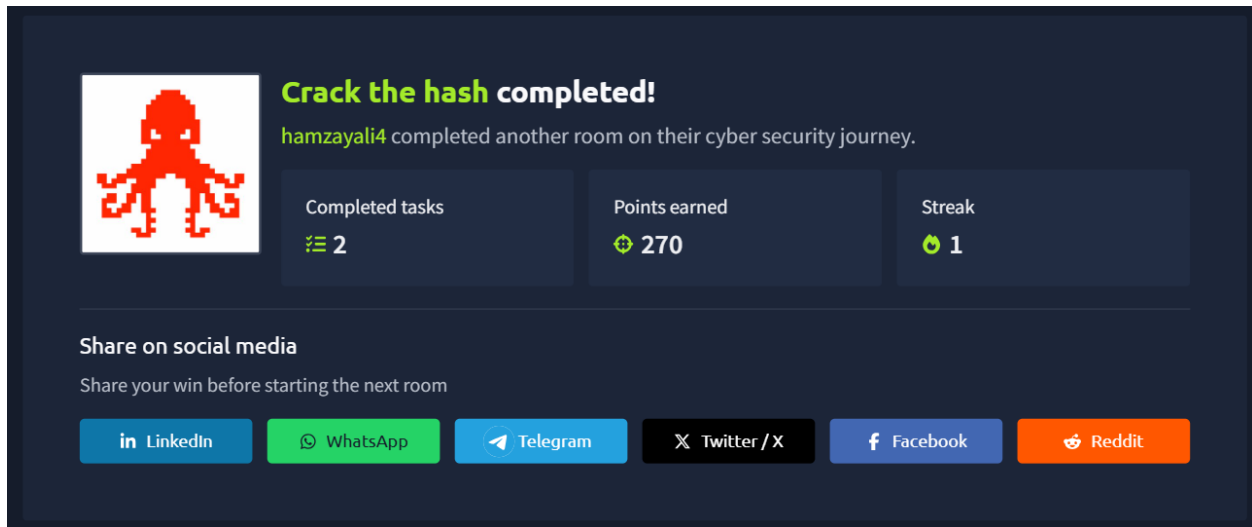
Understand password hash cracking concepts before performing the practical lab.

Description

The TryHackMe "Crack the Hash" room was reviewed to understand hashing algorithms, dictionary attacks, password recovery techniques, and password security best practices.

Screenshot

Figure 1: Crack the Hash – TryHackMe Room



Crack the hash completed!
hamzayali4 completed another room on their cyber security journey.

Completed tasks	Points earned	Streak
2	270	1

Share on social media
Share your win before starting the next room

[in LinkedIn](#) [WhatsApp](#) [Telegram](#) [Twitter / X](#) [Facebook](#) [Reddit](#)

Task 2 – Create a Sample MD5 Hash

Command

```
echo -n "pass" | md5sum
```

Output

```
1a1dc91c907325c69271ddf0c944bc72
```

Explanation

The password **pass** was converted into an MD5 hash using the Linux **md5sum** utility. This generated hash was later used as the target for password recovery.

Screenshot

Figure 2: Generate Sample MD5 Hash

```
(venv)-(hamza@hamza)-[~/Hexa/frontend]
$ echo -n 'pass' | md5sum
Command 'pass' not found, but can be installed with:
sudo apt install pass
Command 'pass' not found, but can be installed with:
sudo apt install pass
6cc7354e2729c2f6f372beaf67a3cb07 -

(venv)-(hamza@hamza)-[~/Hexa/frontend]
$ echo -n 'pass' | shasum
Command 'pass' not found, but can be installed with:
sudo apt install pass
78aaac4c24ea4ce3874618a2a5f9c1b97700aba1 -

(venv)-(hamza@hamza)-[~/Hexa/frontend]
$ echo -n 'pass' | sha256sum
Command 'pass' not found, but can be installed with:
sudo apt install pass
4bdc1e24c2378b0901636c10d71cdc8bb2c29b5ce2239c5b3df94bd571a4493 -
```

Task 3 – Save the Hash into a File

Command

```
echo "1a1dc91c907325c69271ddf0c944bc72" > hash.txt
```

Verification

```
cat hash.txt
```

Explanation

The generated MD5 hash was saved into **hash.txt**, allowing password cracking tools to read it as the target hash.

Screenshot

Figure 3: Save and Verify Hash

```
(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$ echo -n "pass" | md5sum
1a1dc91c907325c69271ddf0c944bc72 -

(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$ cat hash.txt
1a1dc91c907325c69271ddf0c944bc72

(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$
```

Task 4 – Recover Password Using CrackStation

Procedure

1. Open CrackStation.
2. Paste the MD5 hash.
3. Click **Crack Hashes**.
4. Observe the recovered plaintext password.

Input Hash

1a1dc91c907325c69271ddf0c944bc72

Result

Recovered Password

pass

Explanation

CrackStation searched its online database of known password hashes and immediately recovered the plaintext password because the hash already existed in its database.

Screenshot

Figure 4: Password Recovery Using CrackStation

CrackStation Password Hashing Security Defuse Security

Defuse.ca · Twitter

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

```
F09EDCB1FCEFC6DF823DC3505A882655FF77375ED8AA2D1C13F640FCCC2D0C85
1DFECA0C02AE40B8619ECF94819CC1B
$6$aReallyHardSalt$6nKUTqzq.UQqmm0p/T7MPpMbGnZXPMAxi4b3M19be.cf13/qxIf.hs
Gp541BqHhSrHVXg/tpdjS6xeKZAs02
e5d8870e5bdd26602cab8dbe07a942c8669e56d6
e5d8870e5bdd26602cab8dbe07a942c8669e56d6
```

I'm not a robot
This site is exceeding reCAPTCHA Enterprise free quota

Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
F09EDCB1FCEFC6DF823DC3505A882655FF77375ED8AA2D1C13F640FCCC2D0C85	sha256	paule
1DFECA0C02AE40B8619ECF94819CC1B	NTLM	n63umy81kF4i
\$6\$aReallyHardSalt\$6nKUTqzq.UQqmm0p/T7MPpMbGnZXPMAxi4b3M19be.cf13/qxIf.hsGp541BqHhSrHVXg/tpdjS6xeKZAs02	Unknown	Unrecognized hash format.

https://defuse.ca

Task 5 – Perform a Wordlist Attack Using John the Ripper

Command

```
john --format=raw-md5 --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
```

Result

Recovered Password

pass

John the Ripper successfully cracked the MD5 hash using the RockYou wordlist.

Explanation

John the Ripper performed an offline dictionary attack by comparing every password inside the RockYou wordlist against the target MD5 hash until a match was found.

Screenshot

Figure 5: John the Ripper Command

```
(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$ john --format=raw-md5 --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
Created directory: /home/hamza/.john
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=3
Press 'q' or Ctrl-C to abort, almost any other key for status
pass(?)
1g 0:00:00:00 DONE (2026-08-07 12:14) 25.00g/s 902400p/s 902400c/s 902400C/s 051993..iloveme14
Use the "--show --format=Raw-MD5" options to display all of the cracked passwords reliably
Session completed.

(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$ john --show hash.txt
0 password hashes cracked, 2 left

(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$ john --show --format=Raw-MD5 hash.txt
?:pass

1 password hash cracked, 0 left
```

Task 6 – Wordlist Attack Result

Output

Loaded 1 password hash

pass

Session completed

Explanation

The password **pass** was successfully recovered, proving that weak passwords can easily be cracked using commonly available wordlists.

Screenshot

Figure 6: Wordlist Attack Using John the Ripper

```
(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$ john --show hash.txt
0 password hashes cracked, 2 left

(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$ john --show --format=Raw-MD5 hash.txt
?:pass

1 password hash cracked, 0 left

(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
```

Task 7 – Verify the Cracked Password

Command

```
echo -n "pass" | md5sum
```

Verification

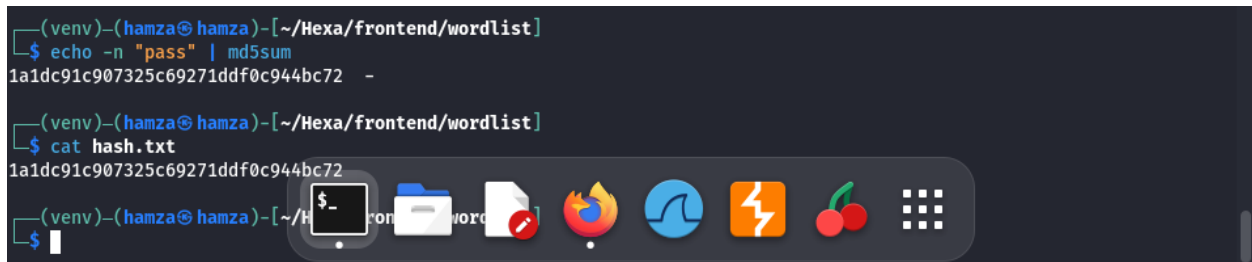
Generated Hash

```
1a1dc91c907325c69271ddf0c944bc72
```

The regenerated hash matched the original hash stored in **hash.txt**, confirming that the recovered password was correct.

Screenshot

Figure 7: Verify Password

A terminal window screenshot showing the verification of a password. The prompt is (venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]. The user enters the command echo -n "pass" | md5sum, and the output is 1a1dc91c907325c69271ddf0c944bc72. The user then enters the command cat hash.txt, and the output is also 1a1dc91c907325c69271ddf0c944bc72. The terminal window has a dark background and a taskbar at the bottom with various application icons.

```
(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$ echo -n "pass" | md5sum
1a1dc91c907325c69271ddf0c944bc72  -

(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$ cat hash.txt
1a1dc91c907325c69271ddf0c944bc72

(venv)-(hamza@hamza)-[~/Hexa/frontend/wordlist]
$
```

Task 8 – Hashcat Reference

Description

The Hashcat Wiki was reviewed to understand supported hash modes, attack methods, and command syntax. Although John the Ripper was used for the practical exercise, Hashcat provides similar functionality for password recovery.

Screenshot

Figure 8: Hashcat Wiki



hashcat

advanced
password
recovery

hashcat Forums Wiki Tools Events

Recent changes Log In Sitemap

Example hashes

If you get a "line length exception" error in hashcat, it is often because the hash mode that you have requested does not match the hash. To verify, you can test your commands against example hashes.

Unless otherwise noted, the password for all example hashes is **hashcat**.

Note also that for many algorithms, when the raw hashes that are components of compound hashes such as `sha1(sha1(pass))`, the hash byte sequence being hashed is the 'hex' (ASCII) form of the hash. The exception is when the suffix `_bin` is present, which indicates that the raw/binary form of the inner hash is what is being hashed by the outer hash function.

NOTE: Some hash types require specific tools to extract the hashes for attack - consult [hash format guidance](#) for your hash type if it's not obvious how to prepare it for attack.

If you are running hashcat from a later GitHub commit or a beta, the latest hash types should always be listed [on GitHub](#).

Table of Contents

- Example hashes
- Generic hash types
- Specific hash types
- Legacy hash types
- Superseded hash types

Generic hash types

Hash-Mode	Hash-Name	Example
0	MD5	8743b52063cd84097a65d1633f5c74f5
10	md5(\$pass.\$salt)	01dfae6e5d4d90d9892622325959afbe:7050461

Difference Between Hashing and Encryption

Hashing	Encryption
One-way process	Two-way process
Cannot be reversed	Can be decrypted using the correct key
Used for password storage	Used to protect confidential data
Used for integrity verification	Used for secure communication
Examples: MD5, SHA-256	Examples: AES, RSA

Results

- Successfully generated an MD5 hash.
- Stored the hash in a text file.
- Recovered the password using CrackStation.
- Performed a successful dictionary attack using John the Ripper.
- Verified the recovered password by regenerating its MD5 hash.
- Compared hashing and encryption concepts.
- Reviewed Hashcat documentation for additional password-cracking techniques.

Conclusion

This lab demonstrated the complete workflow of password hash analysis in a controlled cybersecurity environment. An MD5 hash was generated, saved, and successfully recovered using both an online hash lookup service (CrackStation) and an offline dictionary attack with John the Ripper. The recovered password was verified by generating the same MD5 hash, confirming its correctness. The exercise also highlighted the difference between hashing and encryption and emphasized the importance of using strong passwords and modern password hashing algorithms to defend against dictionary attacks.